Do Now · retrieval
Floating-point addition

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Convert 01011000 0011 to denary. Use an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Q2

Represent 0.375 in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Q3

Convert 11110110 from 8-bit two’s-complement binary to denary. Show signed place values.

Q4

Convert AD_{16} to binary, then logically shift the 8-bit value right once. Give the final denary value.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Align the exponents — Choose the larger exponent. Increase the smaller exponent one step at a time and right-shift that mantissa once for each step.

Core idea 2

Preserve sign and working precision — An arithmetic right shift repeats 0 for a positive mantissa and 1 for a negative mantissa. — Keep low-order guard bits when a shift would otherwise discard a non-zero bit.

Core idea 3

Add using a wider working sum — Sign-extend both aligned mantissas before adding so the calculation cannot wrap into the wrong sign. — CODE:
requested mantissa: 6 bits
working sum: 7 bits

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Normalise and check — If the sum is too large for the mantissa, shift right and increase the exponent. If redundant sign bits remain, shift left and decrease the exponent.

Vocabulary · use the correct technical terms

Exponent alignment — Rewriting operands with a common exponent before adding mantissas.

Arithmetic right shift — A right shift that repeats the sign bit.

Sign extension — Repeating a sign bit when widening a signed working value.

Guard bits — Extra low-order working bits that reduce premature loss of precision.

Independent skill check · apply the lesson

Add -5 and +3 with a 6-bit two's-complement mantissa and 4-bit exponent. Show alignment, signed addition and normalisation.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Add 5 + 1.5

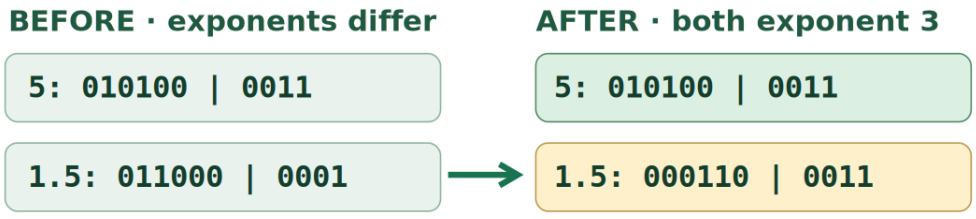
Follow the worked method

1. Use the common exponent 3 — Raise the second exponent from 1 to 3, so shift its mantissa right twice using sign extension. — CODE:
010100 0011
011000 0001 → 000110 0011

2. Return to the requested width — CODE:
0011010
→ 011010 0011 — The mantissa already begins 01 , so the exponent stays at 3.

3. Verify in denary — CODE:
5 + 1.5 = 6.5

Source model · final illustrated step



Arithmetic right shift ×2 · raise exponent 1 → 3

Mantissa | exponent · each number keeps its value

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · solo guided practice

Work ALONE. Try first; read the clues one at a time if you get stuck.

Complete the steps below. Check your answer only on page 8.

YOUR TASK

Build the answer

CLUES · uncover one at a time

Clue 1: Express 6 and 5 with 6-bit mantissas and a common exponent of 3.

Clue 2: Keep an extra sign bit for the working sum before storing a 6-bit mantissa.

Clue 3: If the mantissa is too large, shift it right and increase the exponent.

BUILD YOUR ANSWER · in BLACK PEN

Step 1: Align the two operands.

Step 2: Add the widened signed mantissas.

Step 3: Normalise and state both final fields.

CHECK BEFORE TURNING THE PAGE

Have you shown every step? The worked self-check is at the END on page 8.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Question 1 · OCR OCTOBER 2021 Q2(c) · [5]

Add 011000 0110 and 010100 0100 . Show your working and give your answer in normalised floating-point form using a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Paraphrased exam question; original operands and marks retained. OCR October 2021, Q2(c) .

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Question 4 · ORIGINAL EXAM-STYLE EXPLANATIONS · [5]
Why must exponents match before mantissas are added directly? [2]
What enters from the left when a negative mantissa is shifted right for alignment, and why? [2]
Why can addition lose precision even if the result is within exponent range? [1]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Floating-point mantissas must be aligned to the same exponent before they are added directly.

Use key vocabulary from memory

1. Explain Exponent alignment in your own words.

2. Explain Arithmetic right shift in your own words.

3. Explain Sign extension in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

END SELF-CHECK · only after you have attempted the work

PAGE 5 · Guided We Do solution:

At exponent 3, 0011000 + 0010100 = 0101100 (working width); normalised 010110 0100 = 11.

PAGE 3 · Independent skill check:

-5 + 3 = -2. With common exponent 3, working sum is 1111000 in 7-bit working width. Normalised result: 100000 0001.

Correct any errors in GREEN PEN. Write one improvement.

YOU DO ANSWERS: rhsccs.org/lessons/h446/lesson-006/answers.html